

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: MCN 301

Course Name: DISASTER MANAGEMENT

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

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| 1 | Differentiate between acceptable risk and residual risk. | 3 |
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Acceptable risk, or tolerable risk, is therefore an important subterm; the extent to which a disaster risk is deemed acceptable or tolerable depends on existing social, economic, political, cultural, technical and environmental conditions. In engineering terms, acceptable risk is also used to assess and define the structural and non-structural measures that are needed in order to reduce possible harm to people, property, services and systems to a chosen tolerated level, according to codes or “accepted practice” which are based on known probabilities of hazards and other factors.

Residual risk is the disaster risk that remains even when effective disaster risk reduction measures are in place, and for which emergency response and recovery capacities must be maintained. The presence of residual risk implies a continuing need to develop and support effective capacities for emergency services, preparedness, response and recovery, together with socioeconomic policies such as safety nets and risk transfer mechanisms, as part of a holistic approach.

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| 2 | State the composition of lithosphere? | 3 |
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The earth's crust is not a homogeneous substance; it has different layers of rocks including sedimentary rocks on top, granitic and metamorphic rocks in the middle, and basaltic rocks on the bottom.

The earth's crust also consists of several large dynamic tectonic plates. These tectonic plates move slowly but continuously at an average rate of around 10 cm. Considering this, there was no Atlantic Ocean, and North America and Europe together were one continent 180 million years ago. The Atlantic Ocean came into being because of the drifting apart from the Eurasian and North American plates.

Types of Lithosphere

There are two types of lithosphere:

- Oceanic: Which is associated with oceanic crust and exists in the ocean basins.
- Continental: This is associated with continental crust.

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| 3 | Explain physical vulnerability and ecological vulnerability. | 3 |
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Vulnerability:

- Example: if an area lies near the coast lines, fault lines, unstable hills etc. it makes the area more vulnerable to disasters as compared to an area that is far away from the origin of the disaster.
- Example: Wooden homes are less likely to collapse in an earthquake, but are more vulnerable to fire.

4 Define disaster deficit index. 3

5 Explain psychosocial support related with disaster response. 3

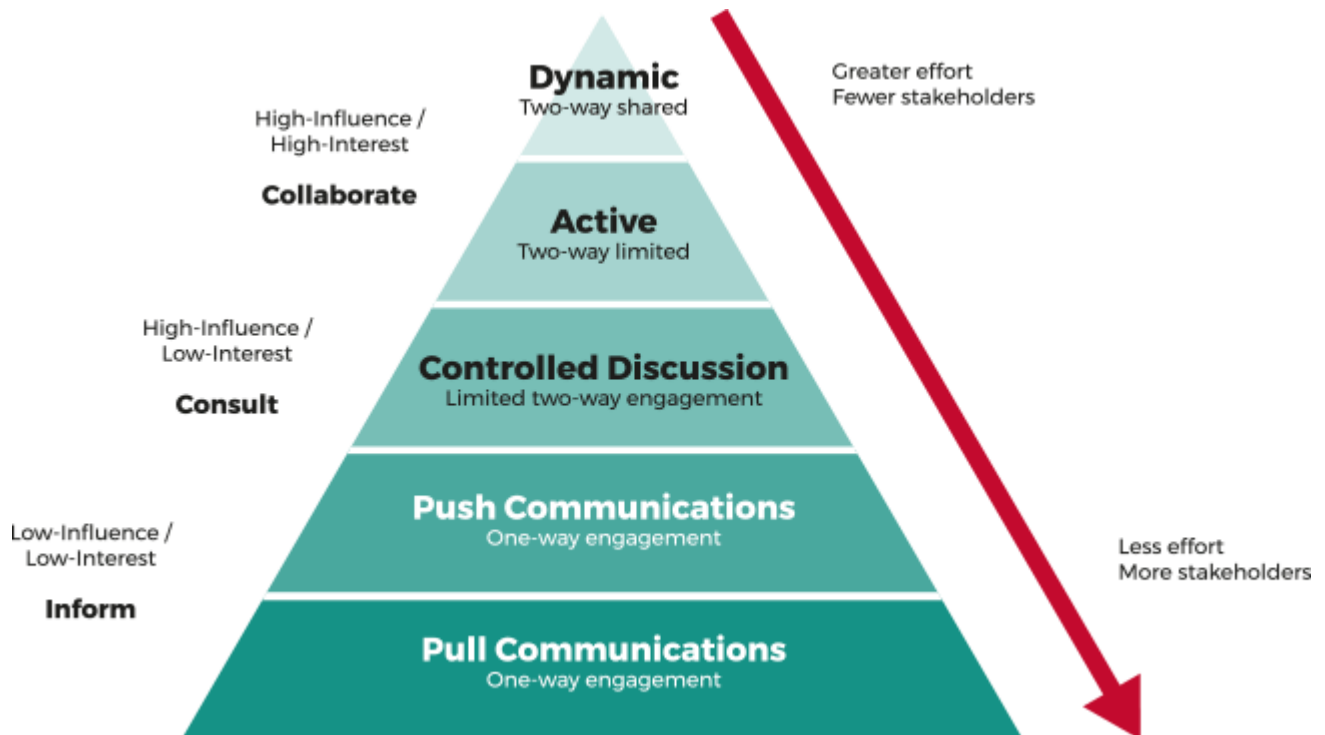
6	List any six public health services required in responding to disasters.	3
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- Human resources management
- Physical resources management
- Victims' management transfer
- Environmental health
- Nutrition Management
- Mental health
- Inter-organizational coordination
- Training

7	Explain pyramid scheme related with participatory stakeholder engagement.	3
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Stakeholder Engagement complements stakeholder Management
Both are needed for Business Value



8 Define the terms counselling and crisis counselling.

3

Crisis Counseling is short term: The idea behind crisis counseling is to help people focus on the stressful situation, treat their immediate needs, and teach them coping skills so they can handle the same or similar situations in the future. Crisis counselors do not replace long-term therapy options.

Man-made disasters have an element of human intent, negligence, or error involving a failure of a man-made system, as opposed to natural disasters resulting from natural hazards. Such man-made disasters are crime, arson, civil disorder, terrorism, war, biological/chemical threat, cyber-attacks, etc.

Examples

Bhopal Gas Tragedy

Morbi Bridge tragedy

Fire at AMRI Hospital, Kolkata

Flyover collapse at Girish Park, Kolkata

Stampede in Allahabad Maha Kumbh mela

1. Training and capacity building: NIDM is responsible for providing training and capacity building programmes to various stakeholders involved in disaster management such as government officials, NGOs, communities, etc. These programmes aim to enhance their knowledge, skills, and capabilities to respond effectively to any kind of disaster.
2. Research and development: NIDM conducts research and studies on various aspects of disaster management, including hazard assessment, vulnerability analysis, risk reduction, and response planning. The findings of these studies are used to develop policies and strategies for disaster management.
3. Disaster management planning and coordination: NIDM is involved in the development of disaster management plans at the national, state, and district levels. It also provides technical assistance and advisory support to various stakeholders for effective coordination of disaster response and recovery operations.
4. Information dissemination: NIDM disseminates information on disaster management through various modes of communication such as seminars, workshops, publications, and digital platforms.

PART B

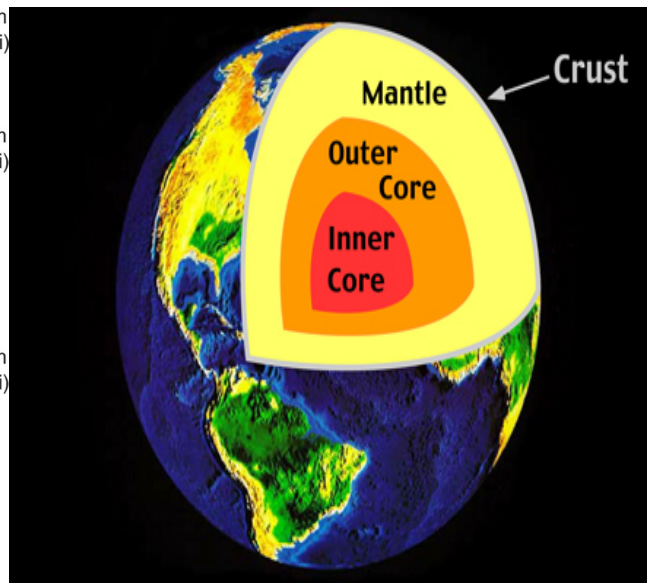
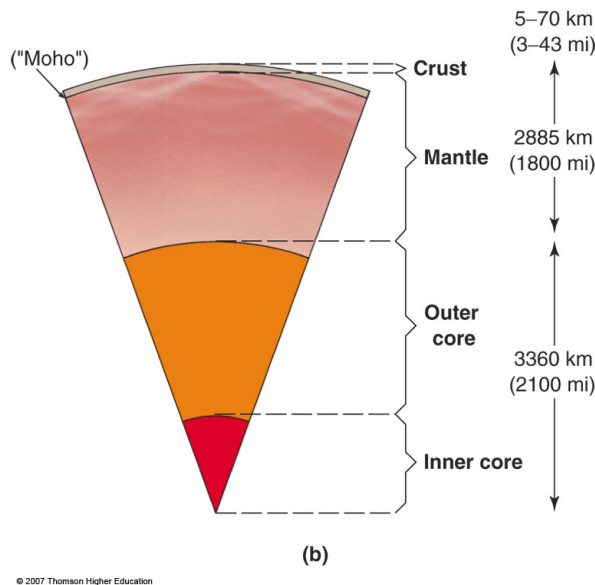
(Answer one full question from each module, each question carries 14 marks)

Module -1

- ☐ The **crust** is the layer that you live on, and it is the most widely studied and understood. The crust is composed of two different rocks. The continental crust is mostly granite. The oceanic crust is basalt.

The crust is only about 3 - 5 miles thick under the oceans (oceanic crust) and about 25 miles thick under the continents (continental crust).

- ❑ The **mantle** is much hotter, has the largest mass, and several layers (uppermost/rigid mantle, asthenosphere, lower mantle).
- ❑ The **outer core and inner core** are even hotter with pressures so great you would be squeezed into a ball smaller than a marble if you were able to go to the center of the Earth



b) State and explain the classification of rocks.

6

There are various types of rocks. They are:

1. Igneous Rocks
2. Sedimentary Rocks
3. Metamorphic Rocks

Igneous Rocks

These rocks are formed by solidification of magma in the interior, or lava on the surface of earth. Igneous rocks are composed of primary minerals, which are predominantly silicates. They sometimes overlap with sedimentary and metamorphic rocks.

Sedimentary Rocks

Sedimentary rocks are formed by the precipitation from solutions, and consolidation of remnants of biotic components like plants and animals. These rocks contain both original primary minerals (Quartz, Mica) and altered as well as newly synthesized secondary minerals (Clay, calcite, gypsum).

Metamorphic Rocks

Also known as Thermal rocks as they are formed from pre-existing rocks (igneous or sedimentary) due to change in the temperature and pressure in solid state is known as metamorphic rocks. These rocks are formed when magma intrudes through pre-existing igneous or sedimentary rocks. All types of pre-existing rocks

could undergo metamorphism. Further, igneous and metamorphic rocks get weathered and form sediments. These sediments get deposited and lithified into sedimentary rocks

- 12 a) State and explain the features of hydrosphere and biosphere in detail. 8
- b) Explain the terms resilience, recovery and early warning systems with respect to disaster risk management. 6

Module -2

- 13 a) Explain the applications of hazard maps. 8
- Hazard maps have various applications that may be broadly captured as in;
- **Spatial planning:** Hazard maps provide a basis for communal and district spatial planning processes
 - **Risk reduction measures:** Hazard maps assist in the localisation and dimensioning of hazard protection measures.
 - **Instruments used in emergency planning:** Hazard maps indicate where the biggest risks arise and the events most likely to occur. This information can be used as a source of orientation in emergency planning.
 - **Raising awareness among the population:** It helps to demonstrate potential risks to the population and to increase awareness of eventual protective measures.
- b) Explain the methods of representing vulnerability. 6

Physical vulnerability

- This refers to the potential losses to physical infrastructure such as roads, bridges, railways, radio and telecommunication mast and other features in the built environment.
- Also includes impacts on the human population in terms of injuries or deaths.

Social vulnerability

- Social vulnerability refers to losses as experienced by people and their social, economic, and political systems
- Vulnerability refers to the extent to which elements of society such as children, the aged, pregnant and lactating women, single parents, physically and mentally challenged, the poor and destitute, social class, caste, ethnicity, gender, family systems, political systems, economic systems and cultural values degrade after being exposed to a hazardous condition.

Economic vulnerability

- This refers to the potential impacts of hazards on economic assets and processes and includes vulnerability of different economic sectors.

Ecological/environmental vulnerability

- This refers to the degree of loss that an ecosystem will sustain to its structure, function and composition as a result of exposure to a hazardous condition.

- 14 a) Explain the method of expressing population risk. 5
- b) List the components of risk assessment. Explain the contemporary approaches to risk assessment. 9

Module -3

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|----|----|--|----|
| 15 | a) | State and explain the types of disaster mitigation measures. | 4 |
| | b) | Identify the factors that determine the nature of disaster response and explain. | 10 |
| 16 | a) | State and explain the types of disaster preparedness. | 6 |
| | b) | Identify the standard operating procedures to be followed during a disaster stage and explain. | 8 |

Module -4

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| 17 | a) | State and explain the basic principles of participatory rural appraisal tools. | 5 |
| | b) | Explain the characteristics of effective crisis counsellors. State the advantages of crisis counselling. | 9 |
| 18 | a) | State and explain the steps for effective communication. | 8 |
| | b) | Identify the barriers to effective communication. | 6 |

Module -5

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|----|----|---|----|
| 19 | a) | List the global targets of Sendai framework and explain. | 7 |
| | b) | Explain the role, composition and responsibilities of National Disaster Management Authority. | 7 |
| 20 | a) | List the guiding principles of Sendai framework and explain. | 10 |
| | b) | State the composition and role of National Executive Committee related with the National Policy on Disaster Management. | 4 |
